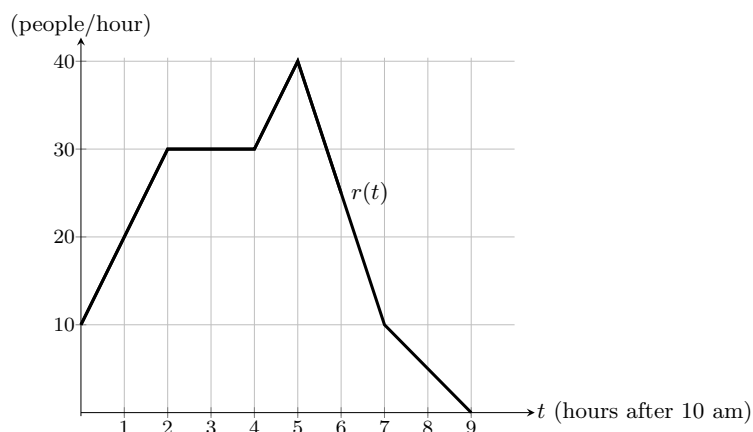


## MATH 1A - PRACTICE PROBLEMS FOR FINAL

1. Your creamy avocado dip (the one that doubles as a face mask from Worksheet 16) has proved to be a giant hit - not only has the stall at the Farmer's Market taken off, you now have a brick-and-mortar store! Your store is open from 10:00 am to 7:00 pm. However, since your store is small, you can only serve 20 people per hour. Let  $r(t)$  be the rate at which people arrive  $t$  hours after 10:00 am, where  $r(t)$  is measured in people per hour. The graph of  $r(t)$  is given to the right.



- (a) If someone arrives at 10:30 am ( $t = 0.5$ ), how long do they have to wait to be served? Explain.

### Solution

At 10:30 am, the rate of people arriving is less than the rate of people being seen into the store, so you shouldn't have to wait at all.

- (b) When does a line start forming?

### Solution

At 11:00 am, as this is when the rate at which people coming to the line (20 ppl/hr) is the same as the rate at which people are getting served.

- (c) Write and evaluate an expression involving an integral that will give you the number of people in line at 2:00 pm ( $t = 4$ ).

### Solution

At 2:00 pm, the number of people in line is

$$\int_1^4 r(t)dt - 3(20) = 85 - 60 = 25$$

- (d) If you arrived exactly at 2:00 pm ( $t = 4$ ), how long would you have to wait in line before you got served? Explain.

### Solution

If you arrived at 2 pm, you would be in a line of 25 people. The rate at which people are

being served is 20 people per hour so you would need to wait a total of

$$\frac{25 \text{ people}}{20 \text{ people per hour}} = \frac{5}{4} \text{ hours}$$

i.e. 1.25 hours (or 1 hour and 15 minutes).

- (e) At approximately what time between 10:00 am ( $t = 0$ ) and 7:00 pm ( $t = 9$ ) is the line the longest? Explain.

#### Solution

As long as  $r(t) > 20$ , we have that the length of the line is increasing. Thus, between 11:00 am and approximately 4:20 pm the line is growing, and after 4:20 pm until 7 pm, the length of the line is decreasing. Thus, the line is longest at approximately 4:20 pm.

- (f) Are there still people waiting in line at 7:00 pm ( $t = 9$ )? How can you tell?

#### Solution

We know that at  $t = 4$  there are 25 people in line, so it is enough to compute

$$25 + \int_4^9 r(t)dt - 5(20)$$

since this represents the number of people arriving in line - the number of people seen from 2 pm to 7 pm. We see this is greater than 0, so there must still be a line.

2. New England is a large producer of maple syrup. Several studies suggest that the amount  $S$  (in liters) of syrup produced by a typical maple tree in one year depends on that year's average winter temperature  $T$  (in  $^{\circ}\text{C}$ ). Suppose that  $S$  and  $T$  are related by the equation

$$4T(1 + 2S)^2 = 30 + \sqrt{4S + T}$$

In 2010, the average winter temperature in Mapleton was  $2^{\circ}\text{C}$ , and the amount of syrup produced by a typical tree was 0.5 liters.

- (a) Find the value of  $\frac{dS}{dT}$  then.

### Solution

We use implicit differentiation:

$$\begin{aligned}\frac{d}{dT}(4T(1 + 2S)^2) &= \frac{d}{dT}(30 + \sqrt{4S + T}) \\ 4(1 + 2S)^2 + 4T(2(1 + 2S)\frac{dS}{dT}) &= \frac{1}{2}(4S + T)^{-1/2}(4\frac{dS}{dT} + 1)\end{aligned}$$

We now can plug in the 2010 info that  $T = 2, S = 0.5$  and simplify to get

$$\begin{aligned}4(2)^2 + 4(2)(2)(1 + 1)(2)\frac{dS}{dT} &= \frac{1}{2}\frac{1}{\sqrt{2 + 2}}(4\frac{dS}{dT} + 1) \\ 16 + 64\frac{dS}{dT} &= \frac{dS}{dT} + \frac{1}{4} \\ 63\frac{dS}{dT} &= \frac{1}{4} - 16 = \frac{-63}{4} \\ \frac{dS}{dT} &= -\frac{1}{4}\end{aligned}$$

- (b) What are the units of  $\frac{dS}{dT}$ ?

### Solution

The units are liters/ $^{\circ}\text{C}$ , as these are the units of  $S$  divided by the units of  $T$ .

- (c) Use your answer in (a) to write a linear function approximating  $S(T)$  (here we are thinking of  $S$  as a function of  $T$ ) near  $T = 2$ . Use your linear function to estimate the amount of syrup produced per tree if the average temperature increased by  $.5^{\circ}\text{C}$ , from  $2^{\circ}\text{C}$  to  $2.5^{\circ}\text{C}$ .

### Solution

Here we want to come up with a tangent line equation, that goes through the point  $(T, S) = (2, \frac{1}{2})$ . Note that in part (a), we already found the slope of the tangent line (i.e.  $dS/dT$ ) which is  $-1/4$  so our equation is

$$S - \frac{1}{2} = -\frac{1}{4}(T - 2)$$

If we want to estimate  $S$  when  $T = 2.5$ , we plug this in to the above to get  $\frac{1}{2} - \frac{1}{4}(2.5 - 2) = \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$  liters.

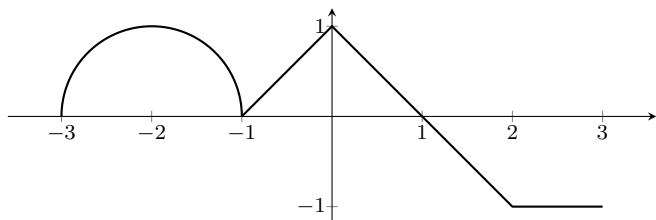
(d) Let  $Q$  be your answer to (a)  $\frac{dS}{dT}$ . Which of the following is the best interpretation of your answer to (a)? Circle your choice; no explanation is necessary.

- A. A typical tree in Mapleton produced about  $Q$  liters of syrup in 2011.
- B. A typical tree in Mapleton produced about  $0.5 + Q$  liters of syrup in 2011.
- C. If the average winter temperature in Mapleton this year is  $2.2^\circ\text{C}$ , then a typical tree should produce about  $0.5 + Q$  liters of syrup.
- D. If the average winter temperature in Mapleton this year is  $1.8^\circ\text{C}$ , then a typical tree should produce about  $0.5 - 0.2Q$  liters of syrup.

### Solution

The correct answer is D.

3. Let  $f$  be the function whose graph is given below, and let  $G(x) = \int_1^x f(t) dt$ .



(a) Evaluate the following, or say that they do not exist.

i.  $G(-1)$

### Solution

$G(-1) = \int_1^{-1} f(t) dt$  Note that  $\int_{-1}^1 f(t) dt$  is the area of a triangle with base 2 and height 1, so has value 1. Since  $G(-1)$  has the bounds switched, this gives  $G(-1) = -1$ .

ii.  $G'(-1)$

### Solution

This is  $f(-1)$  by FTC 1, so 0.

iii.  $G''(1)$

**Solution**

This is  $f'(1)$  (again by FTOC 1), which means we are looking for the slope of the line (of  $f$ , going from  $(0, 1)$  to  $(2, -1)$ ), which is  $-1$ .

iv.  $\lim_{h \rightarrow 0} \frac{G'(-1+h) - G'(-1)}{h}$

**Solution**

This is  $G''(-1)$  (using the limit definition of the derivative), which is  $f'(-1)$  by FTOC 1. Note that since  $f$  has a cusp at  $x = -1$ , it is not differentiable there, so this limit does not exist.

- (b) For what values of  $x$  in  $(-3, 3)$  is  $G(x)$  increasing?

**Solution**

This is asking where  $G'(x) = f(x)$  is positive, which is on  $(-3, 1)$ .

- (c) For what values of  $x$  in  $(-3, 3)$  is  $G(x)$  concave up?

**Solution**

This is asking where  $G''(x)$  is positive, or equivalently, where  $G'(x) = f(x)$  is increasing. This is on the intervals  $(-3, -2)$  and  $(-1, 0)$ .

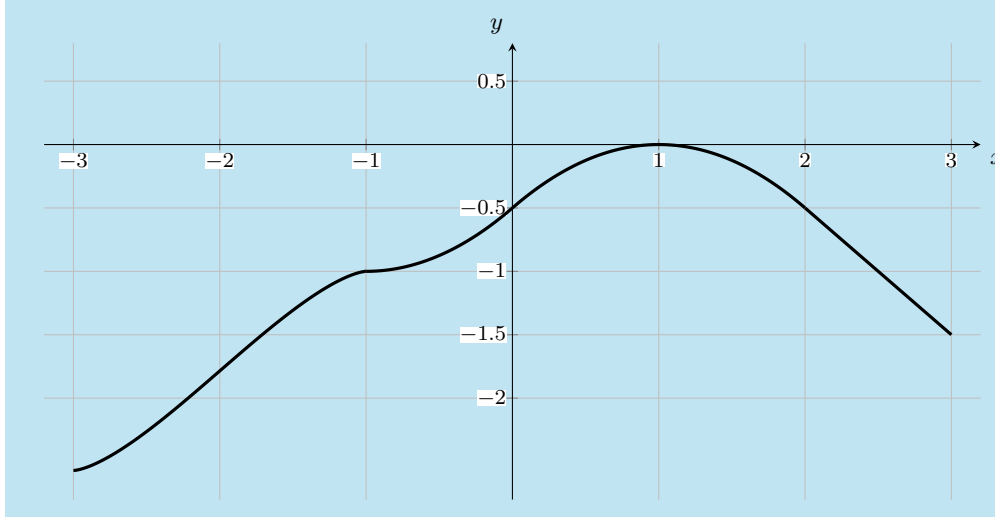
- (d) For what values of  $x$  in  $(-3, 3)$  is  $G(x)$  linear?

**Solution**

This is when  $G'(x) = f(x)$  is constant, or when  $2 < x < 3$ .

- (e) Sketch the graph of  $G$  on  $[-3, 3]$  on the axes below with respect to information from (a)-(d).

### Solution



- (f) Find all  $x$ - and  $y$ -intercepts of  $G$  on  $[-3, 3]$  and label them on the graph above.

### Solution

The  $x$ -intercept is  $(1, 0)$  and the  $y$ -intercept is  $(0, -1/2)$  since  $G(0) = \int_1^0 f(t) dt = -1/2$

4. The Dunder Mifflin Paper Company needs to put in a design bid to get a contract account with Amazon.

The box they are designing is for Amazon's Prime Pantry service. The box must have a rectangular base whose width is twice as long as its length and has a volume of 9 cubic feet. For a competitive bid, they want to minimize the cost of each box by minimizing the surface area. What dimensions of the box should they choose to achieve that? **Make sure to justify that your answer is a minimum.**

### Solution

We start with our goal: minimize surface area. We also note that our volume is constant at 9 cubic feet, which means that

$$V = lwh = 9$$

But since  $w = 2l$  this means that  $V = 2l^2h = 9$ . We will use this to relate our two variables, but first let us come up with an equation for surface area:

$$S = 2l^2(2) + 2lh(2) + lh(2) = 2l^2 + 6lh$$

Ultimately we want to take a derivative of  $S$  so we must rewrite it in one variable. Here it seems easier to use the relationship  $h = \frac{9}{2l^2}$ . Using this we get

$$S = 4l^2 + \frac{27}{l}$$

Thus,

$$\frac{dS}{dl} = 8l - \frac{27}{l^2}$$

Note that in order to find critical points,  $S'(l)$  must be either zero or undefined, but  $S'(l)$  is only undefined when  $l = 0$ , which is not in our domain. Therefore, we set  $S'(l) = 0$  to solve for the critical points:

$$\begin{aligned} 8l &= \frac{27}{l^2} \\ 8l^3 &= 27 \\ l^3 &= \frac{27}{8} \\ l &= \left(\frac{27}{8}\right)^{1/3} = \frac{3}{2} \end{aligned}$$

We could use either a first or second derivative test to check that this is a local min. Taking a second derivative seems a bit easier:

$$S''(l) = 8 + \frac{27(2)}{l^3},$$

which is always positive, so  $S$  is a concave up function and  $l = \frac{3}{2}$  is a local min. Since this is our only critical point (and  $S$  is concave up everywhere), this gives that  $l = \frac{3}{2}$  is a global min. Thus, the dimensions we need to minimize surface area (and thus cost) are  $l = \frac{3}{2}, w = 3, h = 2$ .

5. Compute the following. No need to simplify.

(a)  $\frac{d}{dx} (\pi \ln(1+x^2)\sqrt{x})$

### Solution

From the constant multiple rule, the product rule, and the chain rule, we have

$$\begin{aligned}\frac{d}{dx} (\pi \ln(1+x^2)\sqrt{x}) &= \pi \left( \ln(1+x^2) \frac{d}{dx} \sqrt{x} + \sqrt{x} \frac{d}{dx} \ln(1+x^2) \right) \\ &= \pi \left( \ln(1+x^2) \frac{1}{2\sqrt{x}} + \sqrt{x} \frac{2x}{1+x^2} \right).\end{aligned}$$

(b)  $f'(s)$ , where  $f(s) = s^{1/s}$ .

### Solution

We use logarithmic differentiation: First,

$$\ln(f(s)) = \ln(s^{1/s}) = \frac{\ln(s)}{s}.$$

Taking the derivative of both sides, we find

$$\frac{f'(s)}{f(s)} = \frac{d}{ds} \left( \frac{\ln(s)}{s} \right) = \frac{s \cdot \frac{1}{s} - \ln(s)}{s^2} = \frac{1 - \ln(s)}{s^2}.$$

Multiplying both sides by  $f(s)$ , we find

$$f'(s) = f(s) \cdot \frac{1 - \ln(s)}{s^2} = \left( \frac{1 - \ln(s)}{s^2} \right) s^{1/s}.$$

(c)  $\lim_{x \rightarrow 0^+} \left( \frac{1}{x} - \frac{1}{\sin(x)} \right)$

### Solution

This is an  $\infty - \infty$  indeterminate form, and our trick for these is to put them over a common denominator so that we can apply L'Hôpital's rule:

$$\begin{aligned}\lim_{x \rightarrow 0^+} \left( \frac{1}{x} - \frac{1}{\sin(x)} \right) &= \lim_{x \rightarrow 0^+} \left( \frac{\sin(x) - x}{x \sin(x)} \right) \\ &\stackrel{L'H}{=} \lim_{x \rightarrow 0^+} \left( \frac{\cos(x) - 1}{x \cos(x) + \sin(x)} \right).\end{aligned}$$

This is still  $\frac{0}{0}$  indeterminate, so we apply L'Hôpital's rule again:

$$\begin{aligned}\lim_{x \rightarrow 0^+} \left( \frac{1}{x} - \frac{1}{\sin(x)} \right) &= \lim_{x \rightarrow 0^+} \left( \frac{\cos(x) - 1}{x \cos(x) + \sin(x)} \right) \\ &\stackrel{L'H}{=} \lim_{x \rightarrow 0^+} \left( \frac{-\sin(x)}{-x \sin(x) + \cos(x) + \cos(x)} \right) = \frac{0}{1} = 0.\end{aligned}$$



(d)  $\lim_{x \rightarrow -\frac{\pi}{2}} \frac{\int_{-\frac{\pi}{2}}^x \sin(3t) dt}{x + \frac{\pi}{2}}$

### Solution

This is a  $\frac{0}{0}$  indeterminate form, so we can apply L'Hôpital's rule and the fundamental theorem of calculus:

$$\lim_{x \rightarrow -\frac{\pi}{2}} \frac{\int_{-\frac{\pi}{2}}^x \sin(3t) dt}{x + \frac{\pi}{2}} = \lim_{x \rightarrow -\frac{\pi}{2}} \frac{\sin(3x)}{1} = \sin\left(-\frac{3\pi}{2}\right) = 1$$

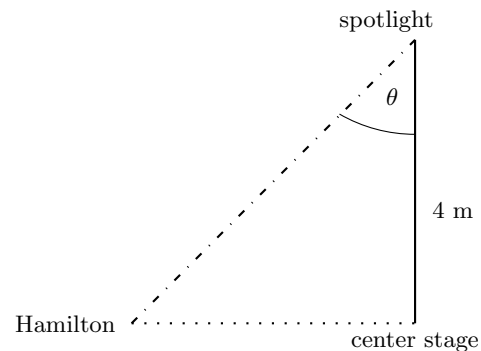
(e)  $\int x^2 \cdot 3^{x^3+1} dx$

### Solution

We make the substitution  $u = x^3 + 1$ , so  $du = 3x^2 dx$ :

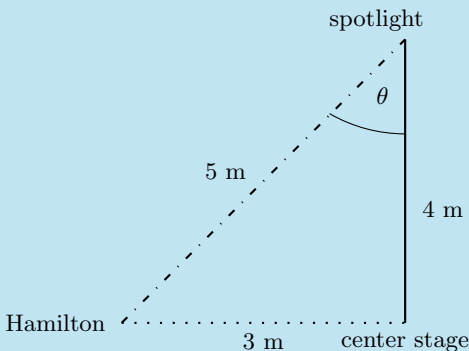
$$\int x^2 3^{x^3+1} dx = \frac{1}{3} \int 3^u du = \frac{1}{3 \ln(3)} 3^u + C = \frac{1}{3 \ln(3)} 3^{x^3+1} + C.$$

6. In preparation for the musical Hamilton, the stage crew is setting up the spotlights. When Hamilton walks left from center stage, he does so at a constant rate of 1.5 meters per second. The spotlight, located 4 meters directly above center stage follows Hamilton as he leaves the stage in a straight line path. When Hamilton is 3 meters to the left of center stage, how quickly is the angle of rotation of the spotlight (marked as  $\theta$  on the sketch) changing as it follows Hamilton?

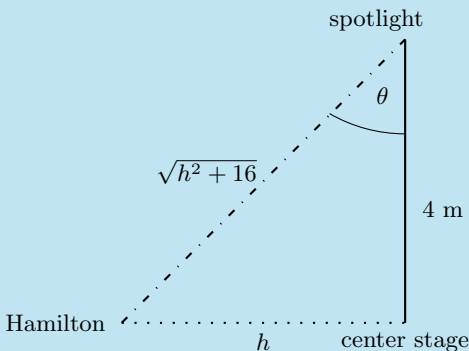


### Solution

Let's make our snapshot picture, filling in the other sides of the triangle with the Pythagorean theorem,



as well as our general picture, writing  $h$  for the distance between Hamilton and center stage (which has the units of meters):



The quantity  $h$  changes in time,  $h = h(t)$ , and we know that  $\frac{dh}{dt} = 1.5$  meters per second. We also know that  $\theta$  changes in time,  $\theta = \theta(t)$ , but that for all time we have

$$\tan(\theta) = \frac{h}{4}.$$

There are a couple of ways to finish the problem from this point. Here are two of them:

- We can rewrite  $\tan(\theta) = \frac{h}{4}$  as  $\theta = \arctan\left(\frac{h}{4}\right)$ , then take the derivative of both sides in time:

$$\frac{d\theta}{dt} = \frac{d}{dt} \arctan\left(\frac{h}{4}\right) = \frac{1}{1 + \left(\frac{h}{4}\right)^2} \cdot \frac{1}{4} \cdot \frac{dh}{dt}.$$

Plugging in  $h = 3$  and  $\frac{dh}{dt} = 1.5 = \frac{3}{2}$ , we obtain

$$\frac{d\theta}{dt} = \frac{1}{1 + \left(\frac{3}{4}\right)^2} \cdot \frac{1}{4} \cdot \frac{3}{2} = \frac{6}{25} \text{ radians per second.}$$

- If we take the derivative of both sides of  $\tan(\theta) = \frac{h}{4}$  in time, we obtain

$$\frac{d\theta}{dt} \sec^2(\theta) = \frac{1}{4} \frac{dh}{dt} = \frac{1}{4} \cdot 1.5 = \frac{3}{8}.$$

Multiplying both sides by  $\cos^2(\theta)$ , we find

$$\frac{d\theta}{dt} = \frac{3}{8} \cos^2(\theta).$$

Now we need to compute  $\cos^2(\theta) = \cos^2(\arctan(3/4))$ . When Hamilton is 3 meters to the left of center stage, from our snapshot picture we see that  $\cos(\theta) = \frac{4}{5}$ , so at that instant we have

$$\frac{d\theta}{dt} = \frac{3}{8} \left(\frac{4}{5}\right)^2 = \frac{6}{25} \text{ radians per second.}$$